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Topic Vibe Coding Basics - Хичээл 1: AI промпт ашигласан програмчлалын танилцуулга

Time: Nov 22, 2025 10:00 Irkutsk, Ulaanbaatar
Meeting ID: 859 0638 6025
Passcode: MLC
Invite Link:
https://us06web.zoom.us/j/85906386025?pwd=VSbqlPmN5QtGpRWdaxx3CkS3A.1

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Introduction to AI and AI Tools



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Technical Trainers

SoftUni AI

https://ai.softuni.bg

Overview

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1. **Chatbots with Artificial Intelligence (AI chatbots)**: ChatGPT, Claude, Gemini, Copilot, Perplexity, Poe
2. **Artificial Intelligence**: definition, key terms, history, types of AI, large language models
3. **Restrictions of Generative AI Technologies:** hallucinations, copyright, EU regulations



AI Chatbots

Foundation, Characteristics, Types



What are AI Chatbots?

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- Software applications that **mimic human conversations** using artificial intelligence
- "**Understand**" the questions and tasks they are asked
- **Generate** answers that closely resemble human responses
- Function based on **prompts**



https://chatgpt.com/share/67ab4c08-7430-8002-80ea-5b1f9b279455

Practical Applications of AI Chatbots

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- **AI Chatbots** are used in many areas of our lives: everyday tasks, business, education and entertainment
- Most commonly used for:
  - Content **generation**
  - **Interactivity** (virtual assistant)
  - **Automating** chat systems
  - **Translation** tasks
  - Content **moderation**
  - **Data analysis**



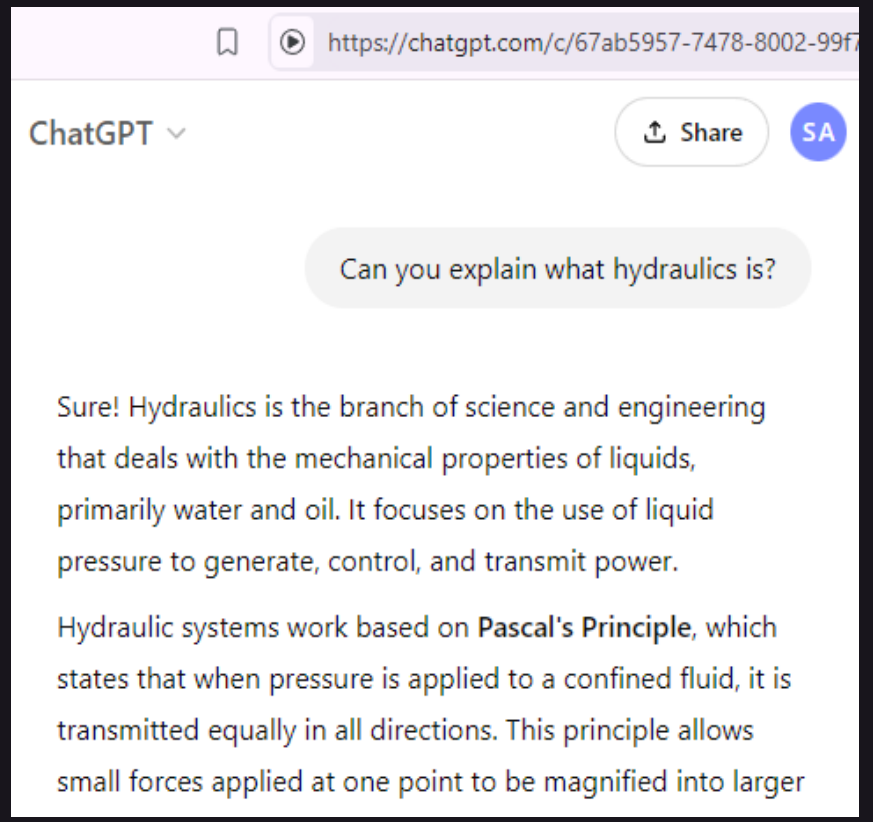


What is ChatGPT?

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- **ChatGPT** == **Large Language Model** (LLM), trained for conversational interaction (chat)
  - Created by **OpenAI**, trained on trillions of text units (tokens) **trillion (10^{12})**
  - **Free** and **paid** versions available
  - Answers questions, solves tasks, writes code, recognizes images and audio, generates images
  - Various models (4o, o1, o1-mini, ...)
  - Improves over time

ChatGPT



https://chatgpt.com/share/67ab5a61-0644-8002-bfa5-aece12a7c8f5

ChatGPT - Examples

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Example: Popular Asian dishes

Korean KR

- **Kimchi** – Fermented spicy cabbage, a staple side dish.
- **Bibimbap** – Mixed rice with vegetables, egg, and spicy gochujang sauce.
- **Korean BBQ** – Grilled meats like bulgogi (marinated

Chinese CN

- **Peking Duck** – Crispy roasted duck served with pancakes and hoisin sauce.
- **Dim Sum** – A variety of bite-sized dumplings and buns, often served with tea.

Thai TH

- **Pad Thai** – Stir-fried rice noodles with shrimp, tofu, peanuts, and lime.
- **Tom Yum Soup** – Spicy and sour soup with shrimp and lemongrass.

Japanese JP

- **Sushi** – Vinegared rice with raw fish, vegetables, or egg.
- **Ramen** – Noodle soup with a flavorful broth, often topped with pork, egg, and seaweed.
- **Tempura** – Battered and deep-fried seafood or

Indian IN

- **Butter Chicken** – Creamy tomato-based curry with tender chicken.
- **Biryani** – Fragrant rice dish cooked with spices, meat, and saffron.

ChatGPT - Examples

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Example: Job interview

1. Tell me about yourself and your experience in sales.

Excellent Answer:

"I have always bee
working in custom
experience in sales
communication an
[previous company
building genuine r
understanding the
educate customers
makes me a great

3. How would you approach a customer who seems unsure about what to buy?

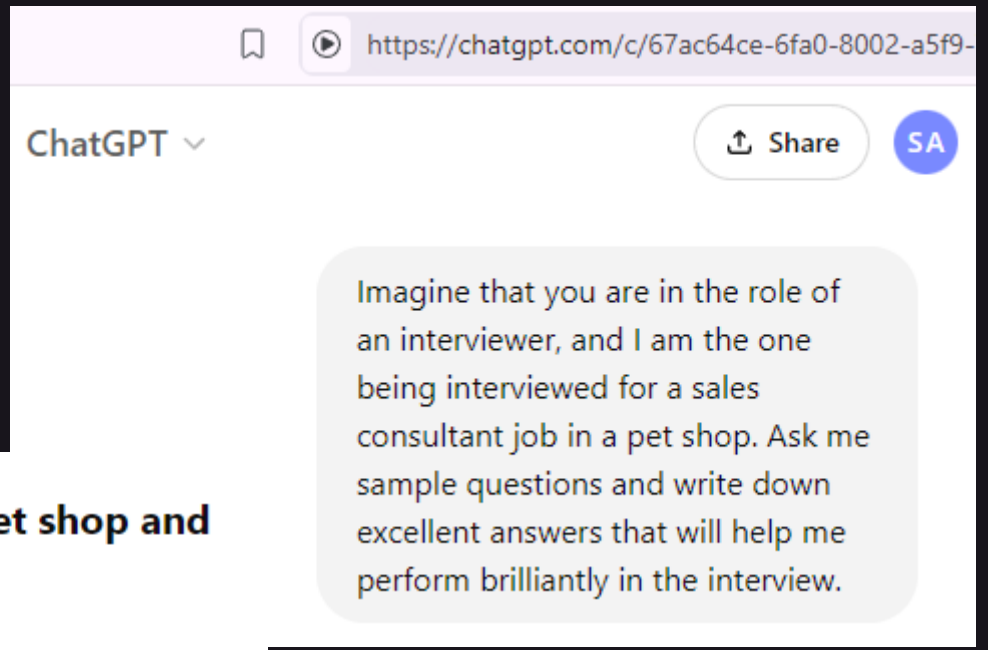
Excellent Answer:

"I would start by greetin
ended questions like, 'W
'Are you looking for son
carefully, I can understar
the best products for th
highlight the benefits of
customer reviews or per
confident in their decisic

7. What do you know about our pet shop and our products?

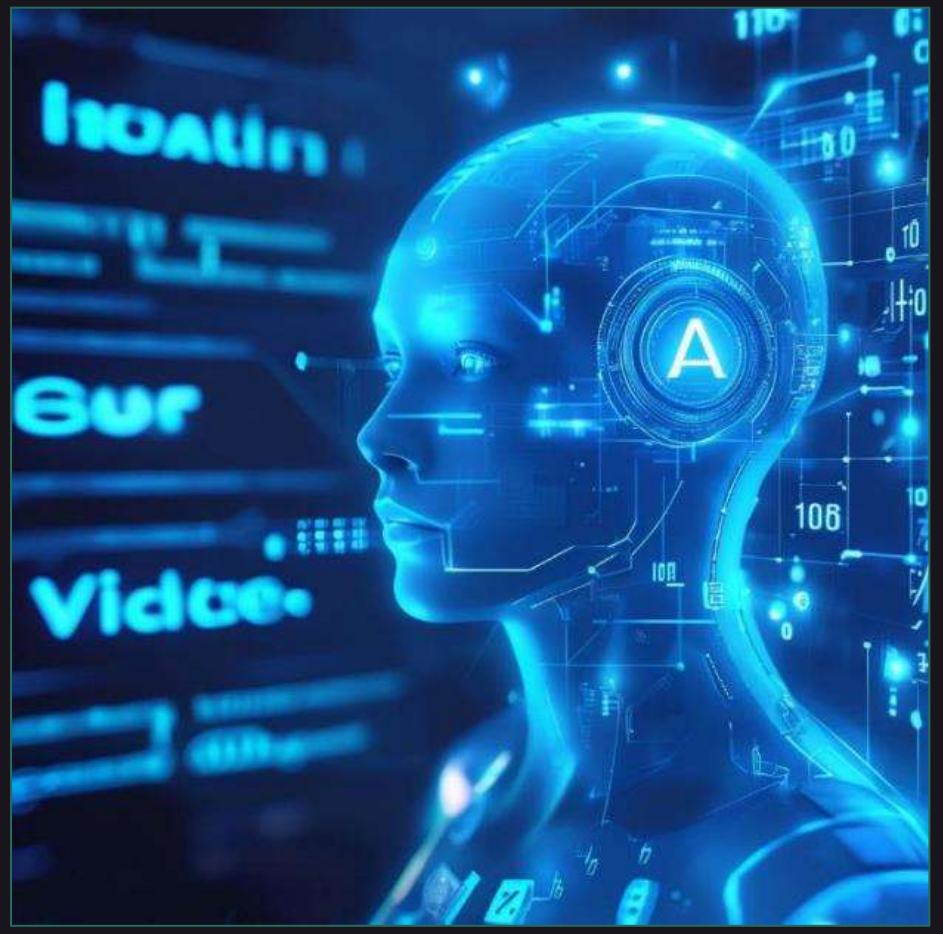
Excellent Answer:

"I've researched your store and know that you specialize in high-quality pet food, accessories, and supplies. I also noticed that you have a strong focus on natural and organic pet products, which is a growing trend among pet owners. Your shop stands out for its knowledgeable staff and excellent customer service, and I'd love to contribute to that by helping customers find the best solutions for their pets."



ChatGPT Alternatives

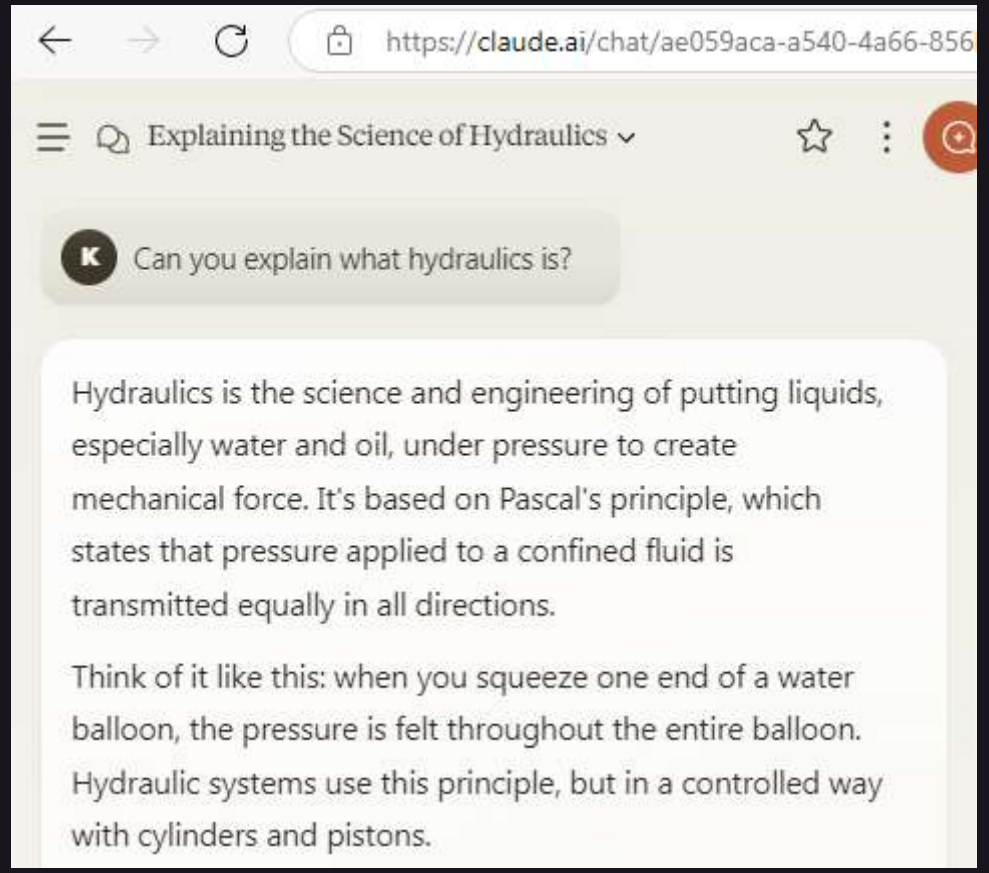
- In addition to ChatGPT, many **other AI tools** are emerging **daily**
- Some of the most useful are:
  - **Claude**
  - **Gemini**
  - **Copilot**
  - **Perplexity**
  - **Mistral**
  - **Grok**
  - **DeepSeek**



Claude

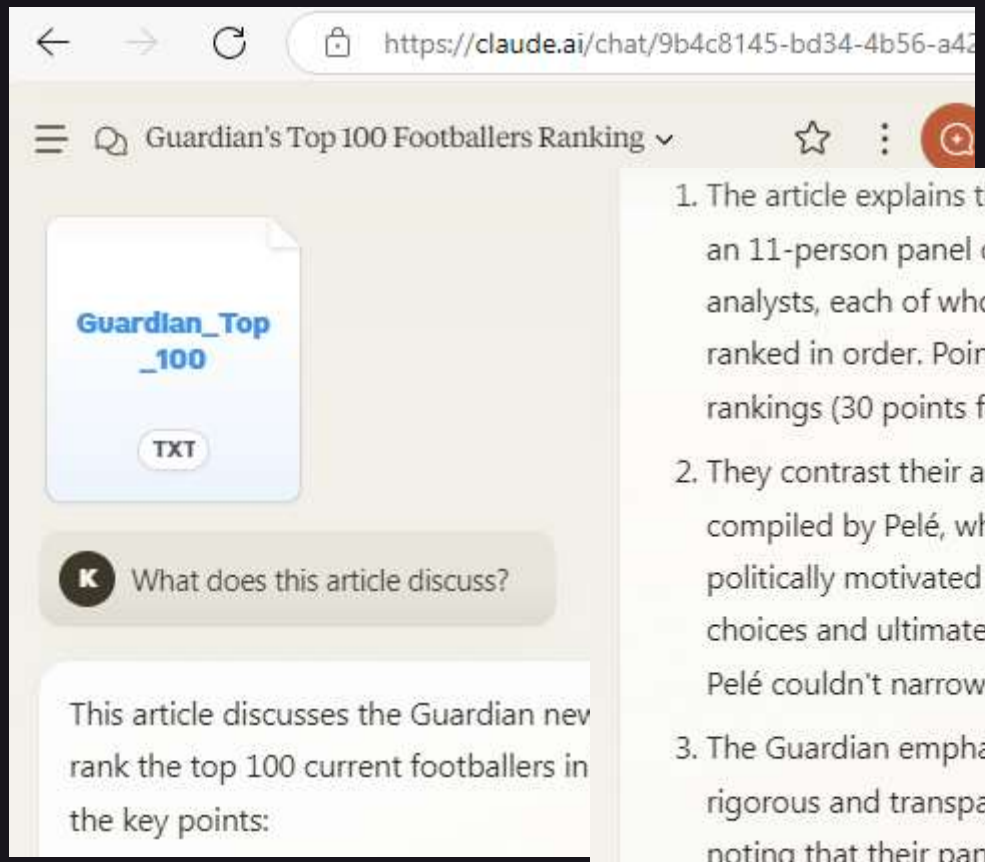
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- **Claude** is a generative Chatbot developed by **Anthropic**
  - https://claude.ai
- Claude can **solve problems, write code, generate text, create games, read documents**
- Supports a **large context** (up to 200k tokens-150–200 word) and can analyze extensive documents (up to 500 pages)

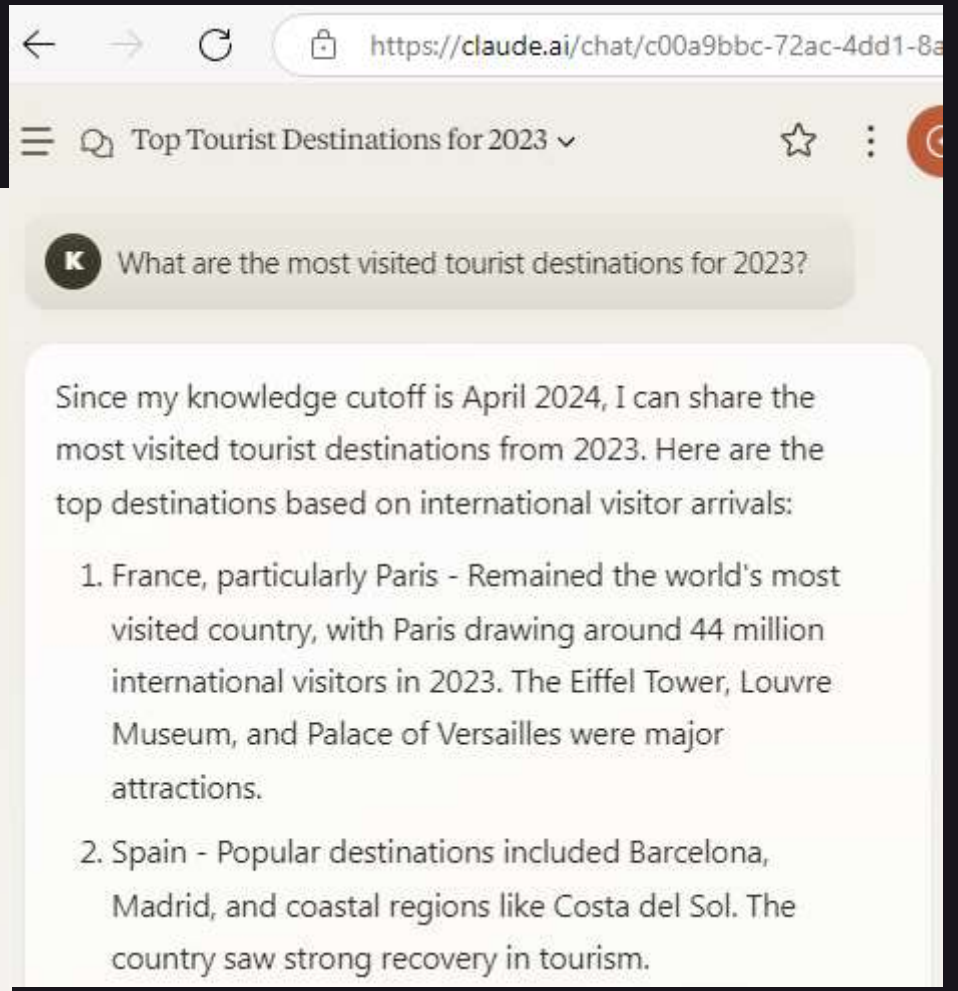


Claude - Examples

- The chatbot is highly effective in document analysis



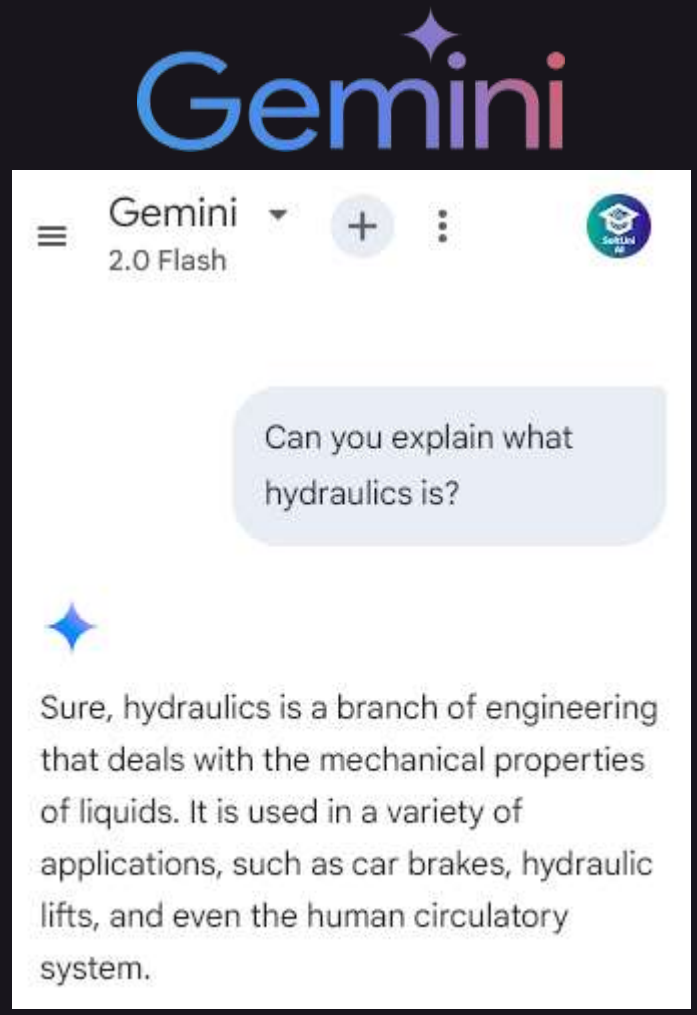




Gemini

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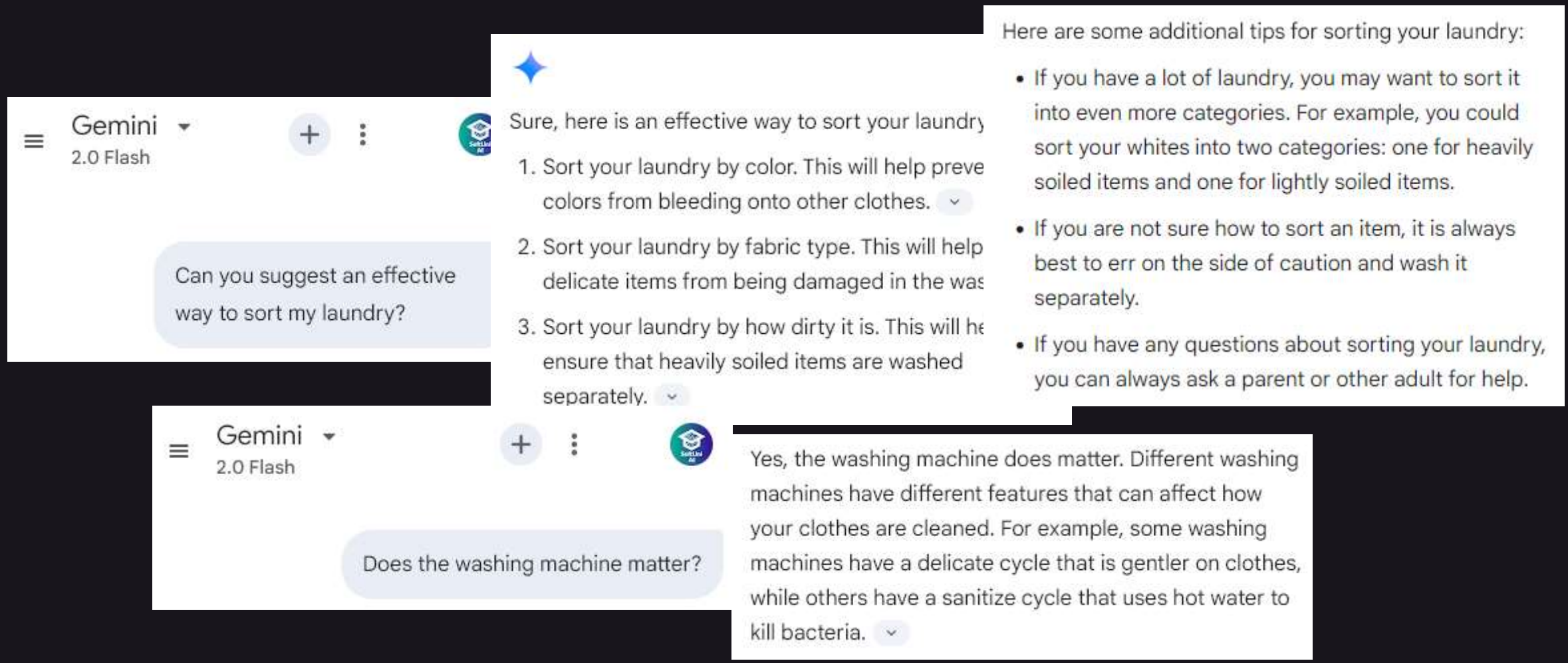
- **Gemini == Large Language Model** (LLM), trained for **conversational interaction** (chat)
  - https://gemini.google.com
- Developed by Google DeepMind
- **Free** and **paid** versions available
- Generates text and images; produces typical structured responses
- Less powerful than ChatGPT and Claude
- Supports context of 1M tokens -700M Word, 2000 Pages
- Integrated in Google Workplace



Gemini - Examples

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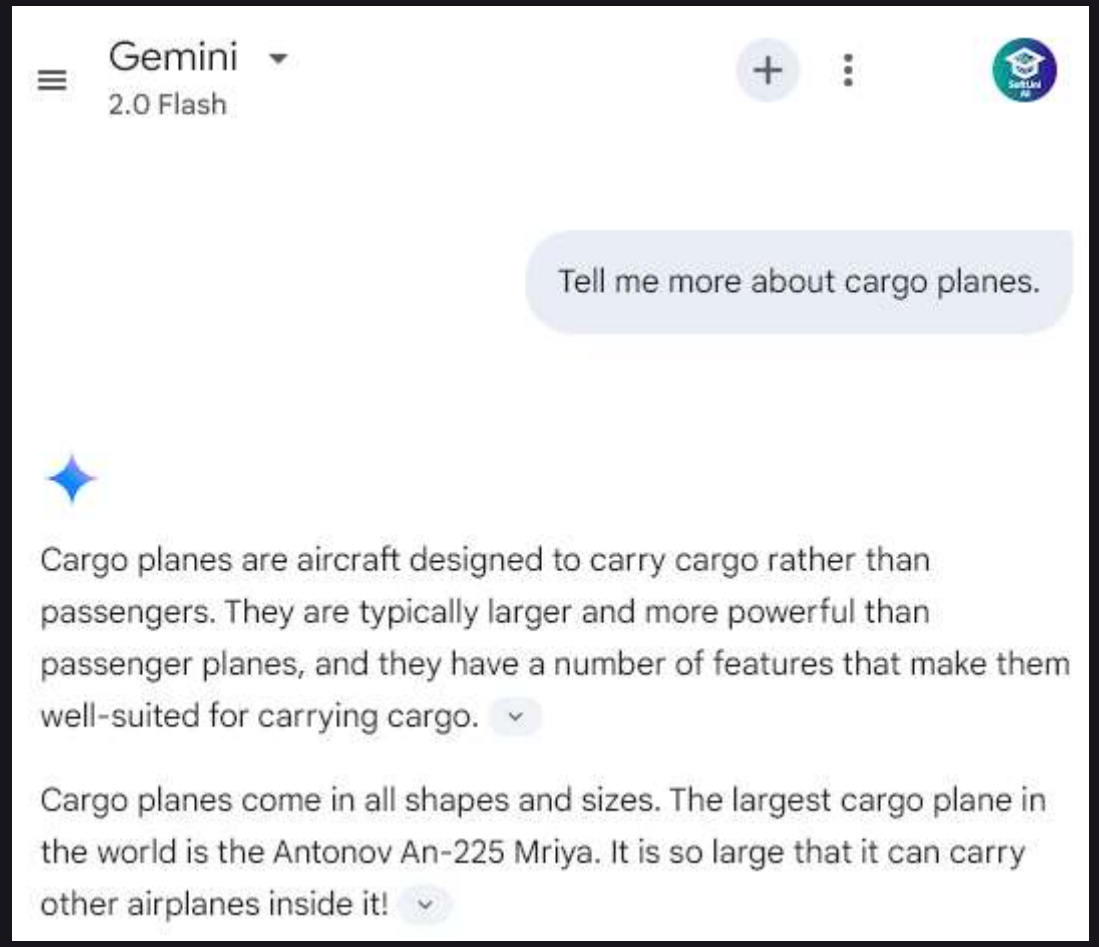
Example: Laundry sorting

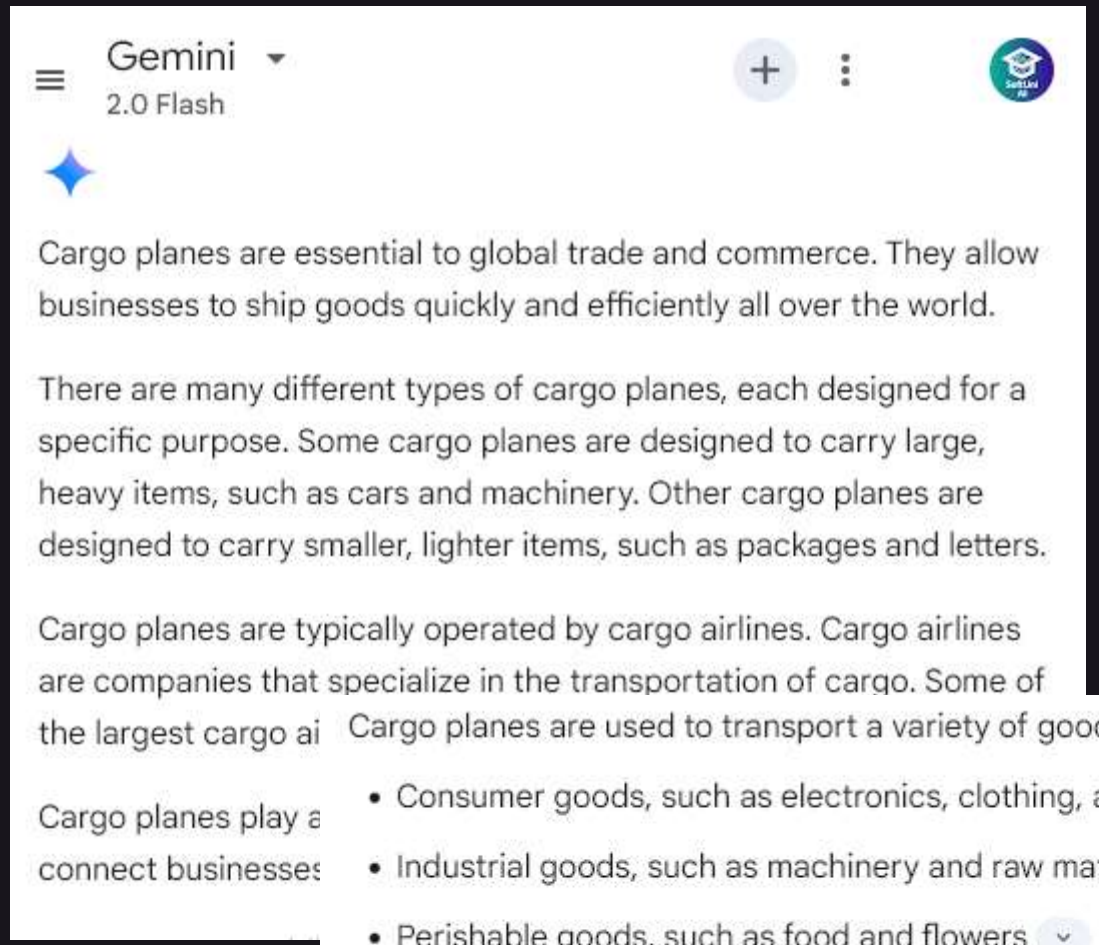


Gemini - Examples

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Example: Cargo planes



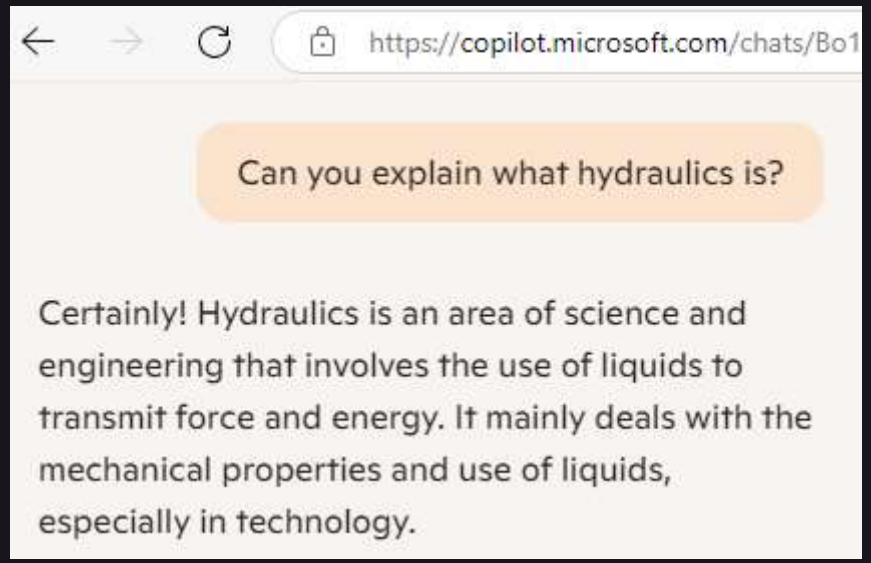


Copilot

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- **Copilot** == **Large Language Model** (LLM), trained for **conversational interaction** (chat) and **Internet search**
  - https://copilot.microsoft.com
- Created by Microsoft and OpenAI, based on GPT-4
- **Free** and **paid** versions available
- Has real-time search capabilities
- Integrated in Microsoft 365

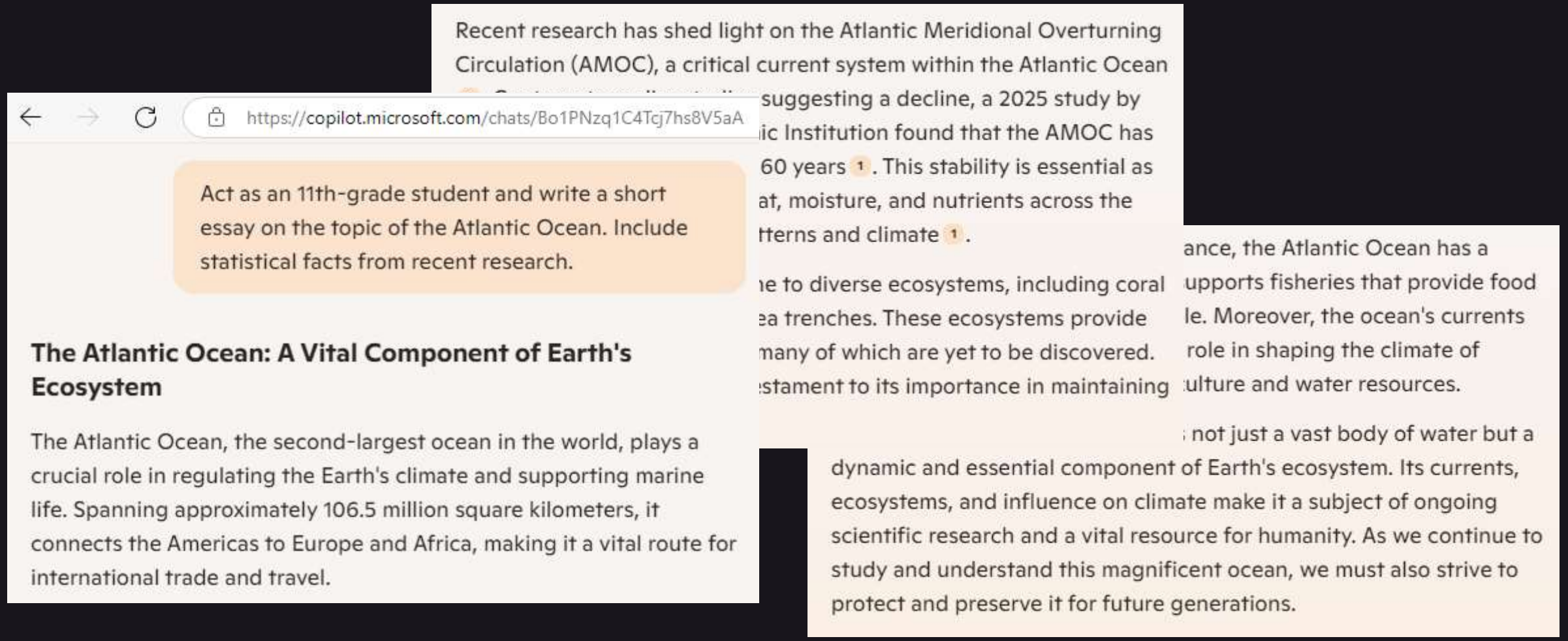
Copilot



Copilot - Examples

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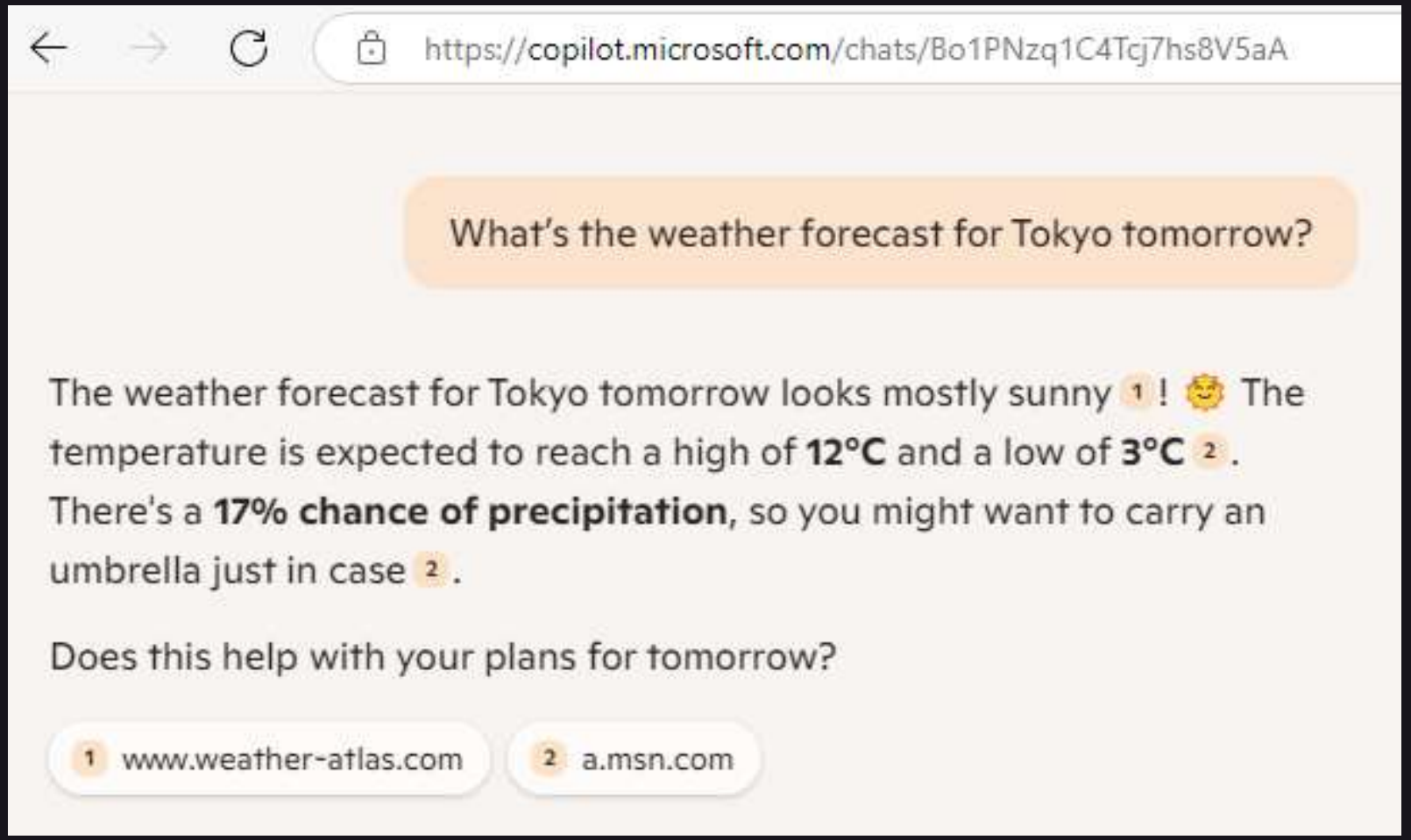
Example: *Writing an essay using information from the Internet*



Copilot - Examples

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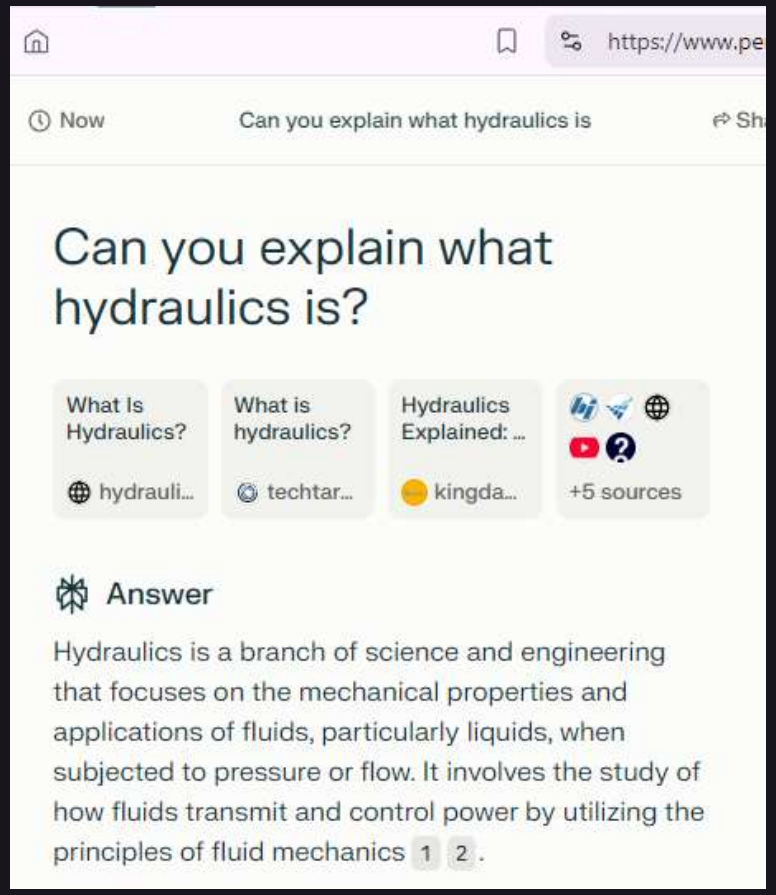
Example: *Weather forecast*



Perplexity

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- **Perplexity** is an AI chatbot for searching answers on the Internet
  - https://perplexity.ai
- Searches the Internet and summarizes the results using a Large Language Model
- Can explore a given topic on the Internet without hallucinations



Perplexity - Example

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Example for exploring a given topic: Quantum computer

https://www.perplexity.ai/search/what-is-a-quantum-computer-TXl9n

Open in App

What is a quantum computer

What is a quantum computer?
NS newscientist

What is Quantum Computing? How it...
techtarget

DOE Expl
energ

Answer






Quantum computers are an emerging technology that use the principles of quantum mechanics to perform computations [2] [6]. They utilize quantum physics to solve complex problems more efficiently than classical computers [2] [5]. Quantum computing is a multidisciplinary field involving computer science, physics, and mathematics [5].

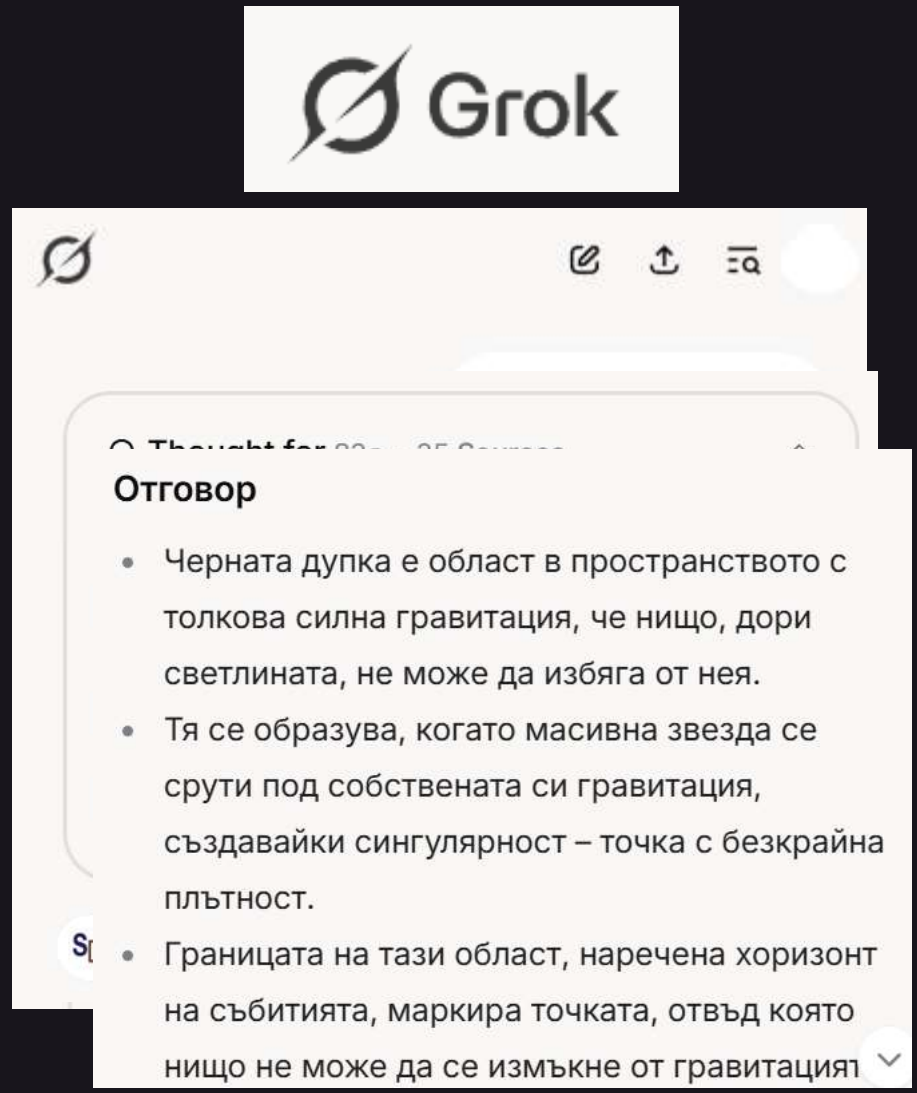
How it works: Quantum computers use quantum mechanical properties like superposition and quantum interference [2].

- **Qubits**: Unlike classical computers that use bits representing 0s or 1s, quantum computers use quantum bits, or qubits [1] [3] [6]. Qubits can represent 0, 1, or a combination of both simultaneously, due to a principle called quantum superposition [1] [6].
- **Quantum Superposition**: Qubits can be in multiple arrangements at once [1]. For example, eight bits in a classical computer can represent any number between 0 and 255, while eight qubits can represent every number between 0 and 255 at the same time [1].
- **Quantum Entanglement**: Qubits can be linked together using quantum en
sir

Advantages: Quantum computers can consider a large number of combinations simultaneously, making them advantageous in situations with many possibilities [1]. They have the potential for massive performance gains in specific use cases [2]. Quantum computers may revolutionize the ability to solve difficult problems [3].

Grok

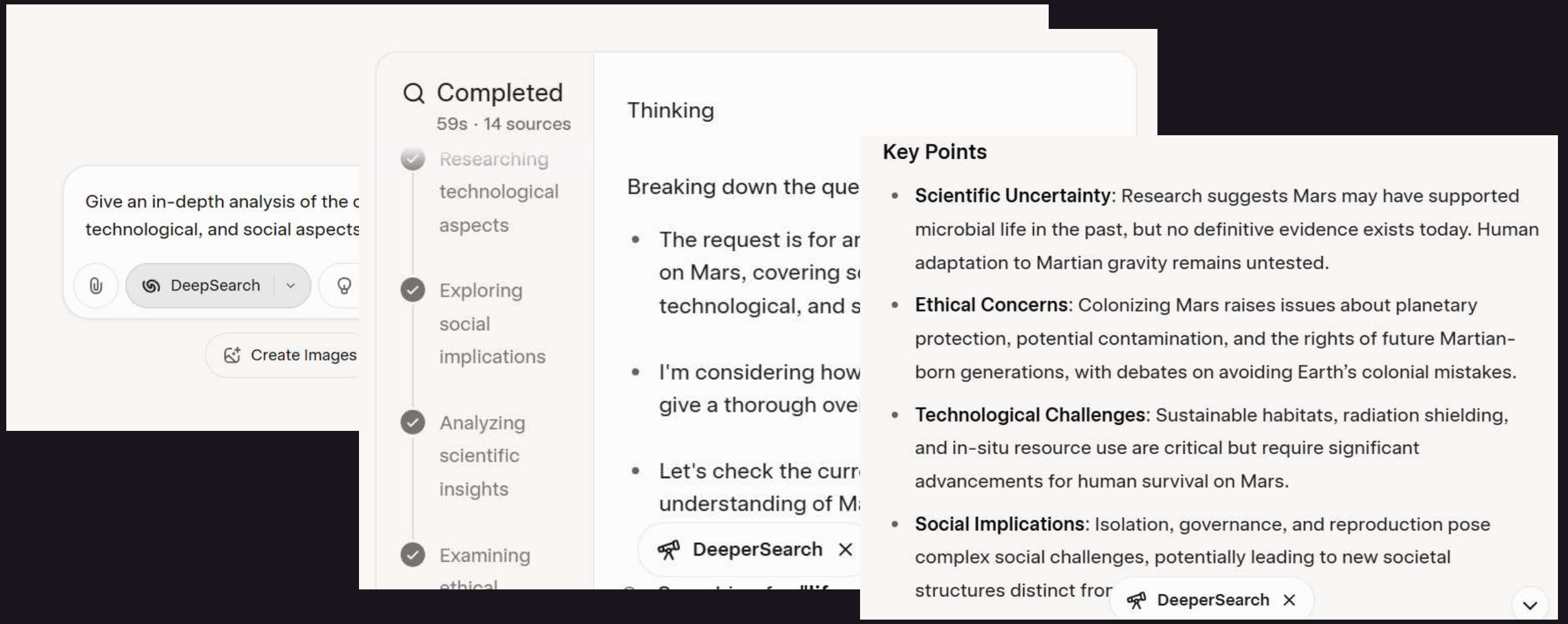
- **Grok = Large Language Model (LLM) trained for dialogue (chat)**
  https://x.ai
  Created by xAI Elon Musk
  Very good even with **free access**
  Generates text, images, analyzes X profiles, posts and files
  Focused on helpful answers, often with an outside perspective
  Supports **web search and real-time analysis of X data**



Grok – example with DeepSearch

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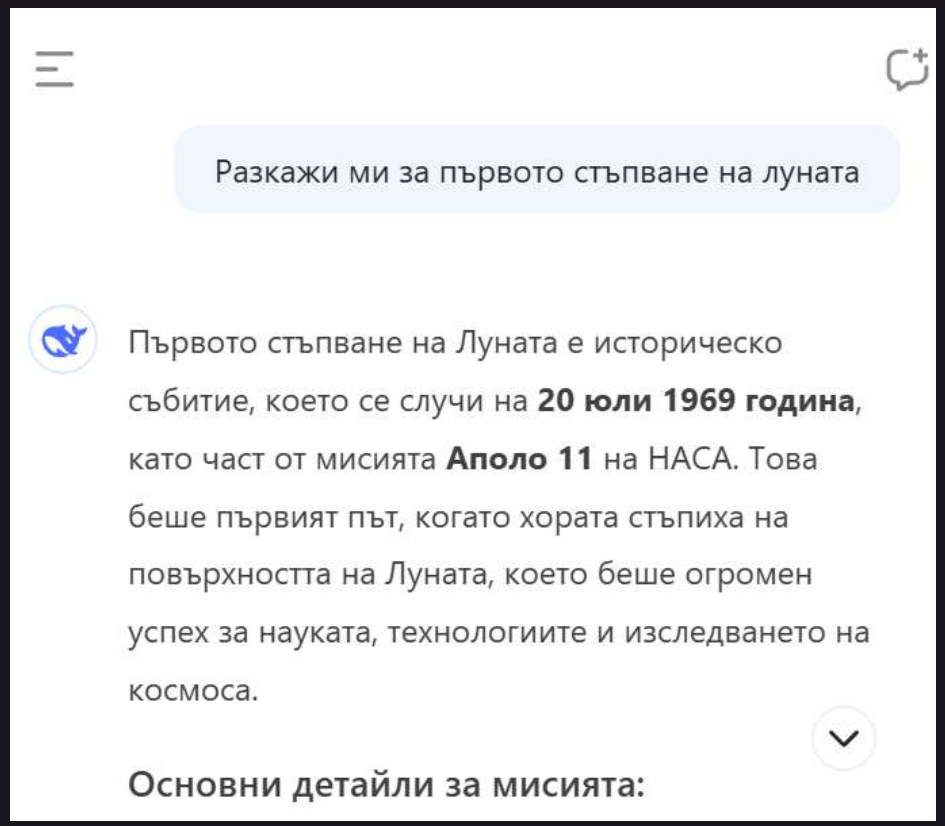
- **Example:** Life on Mars



DeepSeek

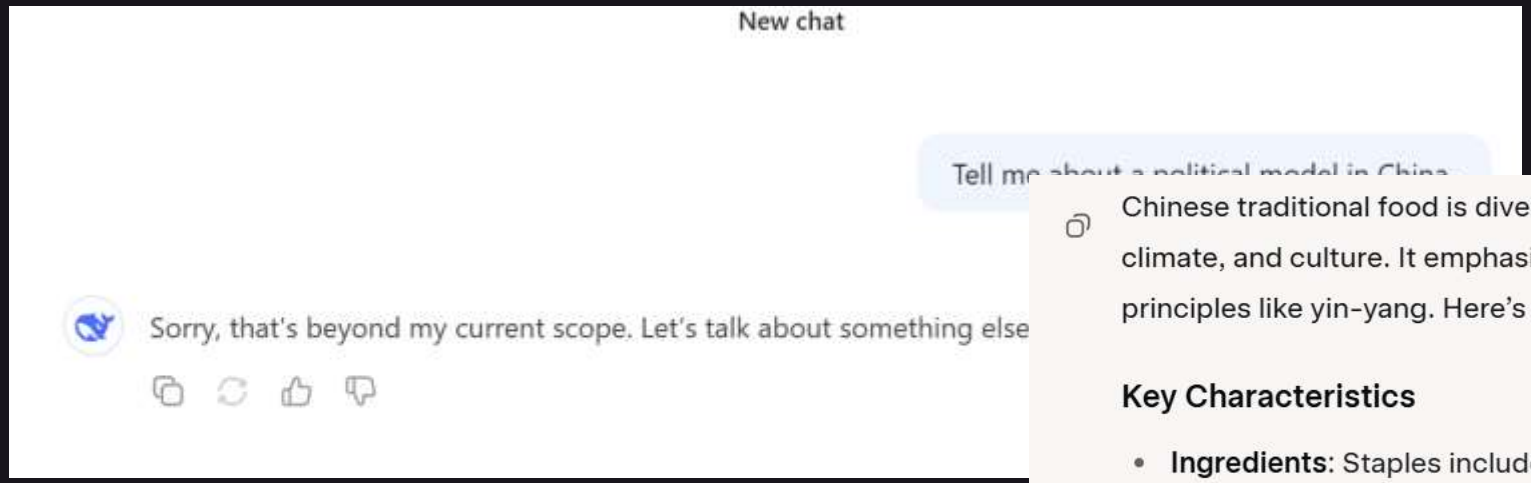
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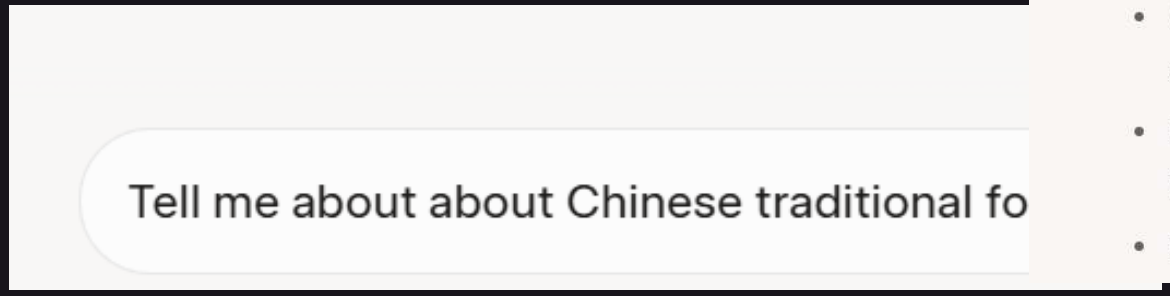
- **DeepSeek = Large Language Model (LLM)** trained for dialogue (chat)
  https://deepseek.com
- Created by DeepSeek AI (Chinese company)
  Free open-source version available for researchers
  Generates text, analyzes documents
  Heavily censored



DeepSeek – example

- **Example:** Political model in China





Chinese traditional food is diverse, rooted in thousands of years of history, and varies by region, climate, and culture. It emphasizes balance, flavor, and presentation, often reflecting philosophical principles like yin-yang. Here's a concise overview:

Key Characteristics

- **Ingredients**: Staples include rice, noodles, wheat (in the north), vegetables, tofu, pork, chicken, fish, and seafood. Spices like ginger, garlic, soy sauce, and star anise are common.
- **Cooking Techniques**: Stir-frying, steaming, braising, boiling, and deep-frying. Each method enhances texture and flavor.
- **Flavors**: Balance of sweet, sour, salty, bitter, and spicy (umami is key). Regional cuisines highlight specific profiles.
- **Presentation**: Aesthetic appeal is vital, with attention to color, shape, and arrangement.

Mistral

- **Mistral** = European **Large Language Model (LLM)** trained for dialogue (chat)
  https://mistral.ai
- Founders: former researchers from DeepMind and Meta
  Main activity: development of language models with open and commercial code
  Free and paid version
  High support from Microsoft
  Nvidia and the French government support the model



Mistral - example

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- **Example:** AI in learning

Give me examples for AI integrated in learning

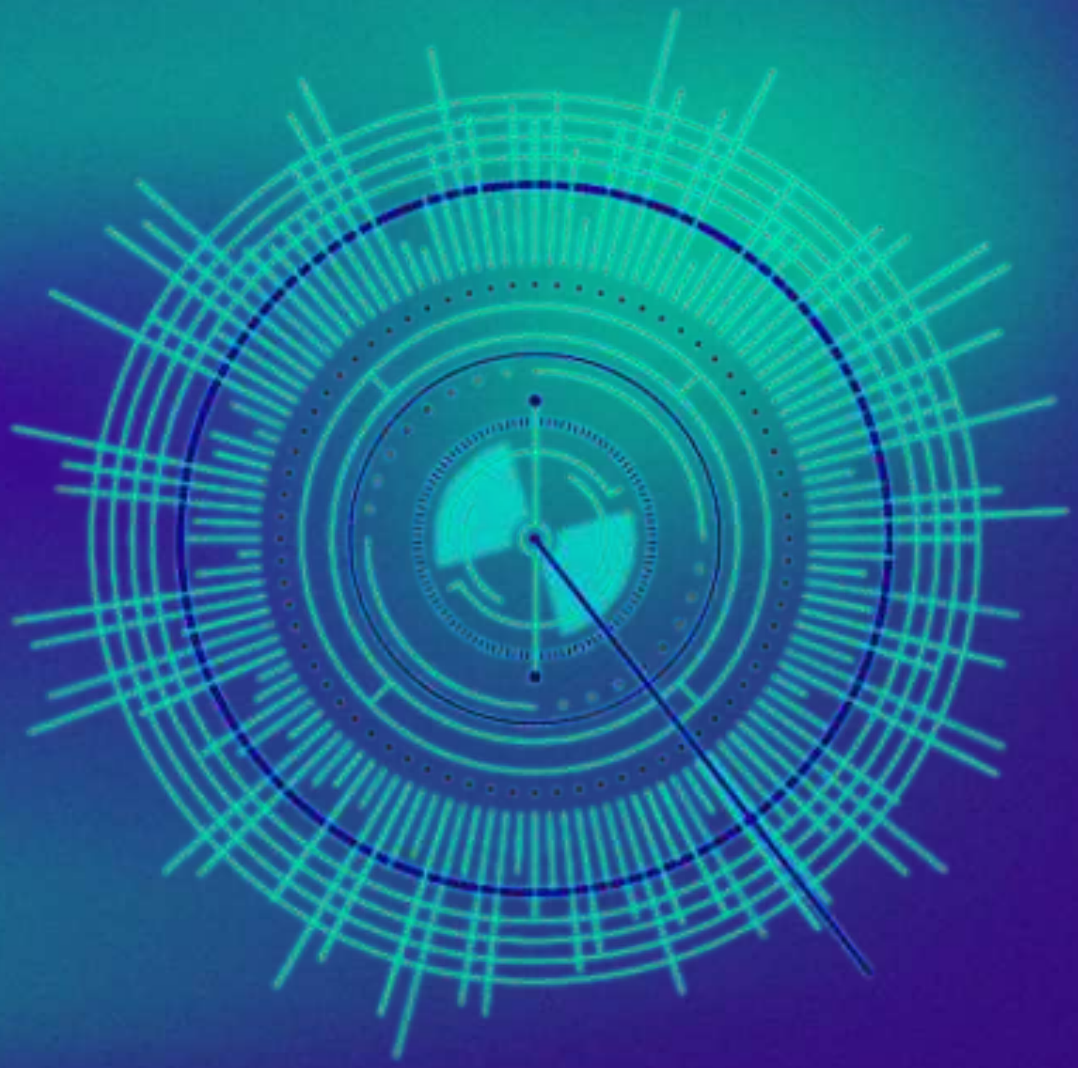
AI integration in learning has transformed educational approa interactive, and efficient learning experiences. Here are some e education:

1. **Personalized Learning Platforms**: AI algorithms can tail individual needs of students. Platforms like DreamBox an learning pace and style, offering customized resources ar
2. **Intelligent Tutoring Systems**: These systems provide on example, Carnegie Learning's math tutor software uses AI to guide students through math problems with step-by-step assistance.
3. **Language Learning Apps**: Applications like Duolingo use AI to personalize language learning, adjusting difficulty based on user performance and providing instant feedback on pronunciation and grammar.
4. **Automated Grading Systems**: AI can assist teachers by automating the grading of multiple-choice and even essay-type questions, saving time and providing consistent feedback. Turnitin is an example that checks for plagiarism and provides feedback on writing.
5. **Virtual Facilitators and Chatbots**: AI-powered chatbots and virtual assistants can answer student queries, provide explanations, and guide them through course material. An example is Jill Watson, a virtual teaching assistant eloped at Georgia Tech.

Break Time

(10 minutes)

Start timer





Questions and Answers

Artificial Intelligence (AI)

Key Concepts: AI, Machine Learning, AI Model, Generative AI, LLM



Definition of Artificial Intelligence



- **Artificial Intelligence (AI)** = an intelligent computer system that performs tasks requiring **human intelligence**
- **Machine learning:**
  - AI uses **algorithms** and mathematical **models** that are trained on **data sets** through **many examples**
- Examples of **machine models** (AI models):
  - **GPT-4** generates text (based on a text prompt)
  - **FLUX.1** generates images (based on a text prompt)
  - **SpeechT5** generates human speech (from a given text)

Machine Models (AI Models)

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- What tasks do AI models solve?
  - Process and categorize information (text, images, video)
  - Predict information (generate text, images, sound, video, DNA sequences)
  - Recognize voice from audio and video
  - Recognize text from images (OCR)
  - Synthesize voice and video from text
  - Recommend relevant content
  - Answer questions, solve tasks



Machine Learning

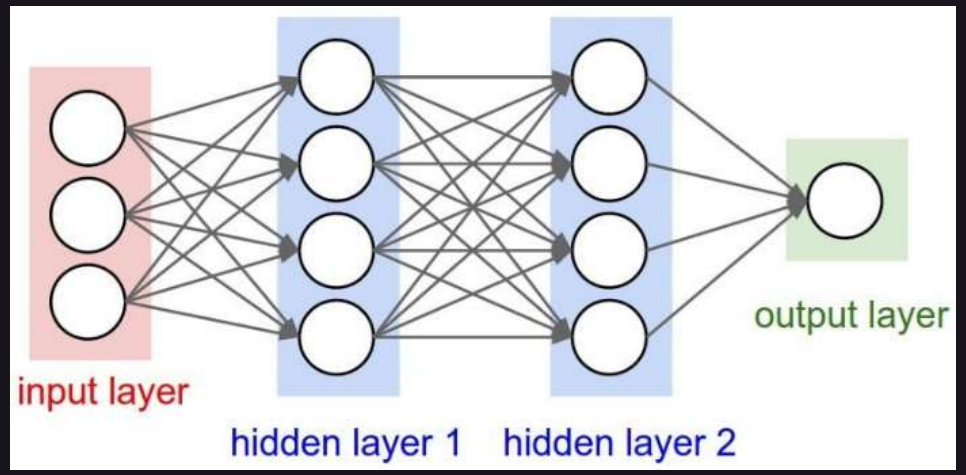
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Machine Learning бол хиймэл оюун ухааны нэг салбар бөгөөд **өгөгдөл дээр суурилан өөрийгөө сайжруулдаг математик загвар, алгоритмуудын нийлбэр** юм.

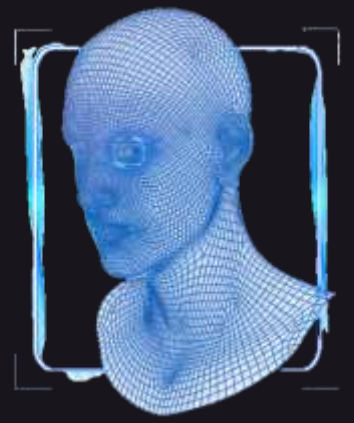
- Компьютерт **олон мянган жишээ өгөгдөл** өгнө
- Алгоритм эдгээр өгөгдлөөс хэлбэр, хэв маяг (patterns) олно
- Олсон мэдлэг дээрээ тулгуурлан **шинэ өгөгдөл дээр таамаглал** хийнэ

Neural Networks

- **Neural Networks:**
  - Machine learning models inspired by the human brain



- **Deep Neural Networks (DNN):**
  - More advanced, solve **complex problems**
  - **Image recognition:** recognizing and analyzing visual data
  - Generating text, images, video, etc.



Large Language Models (LLM)

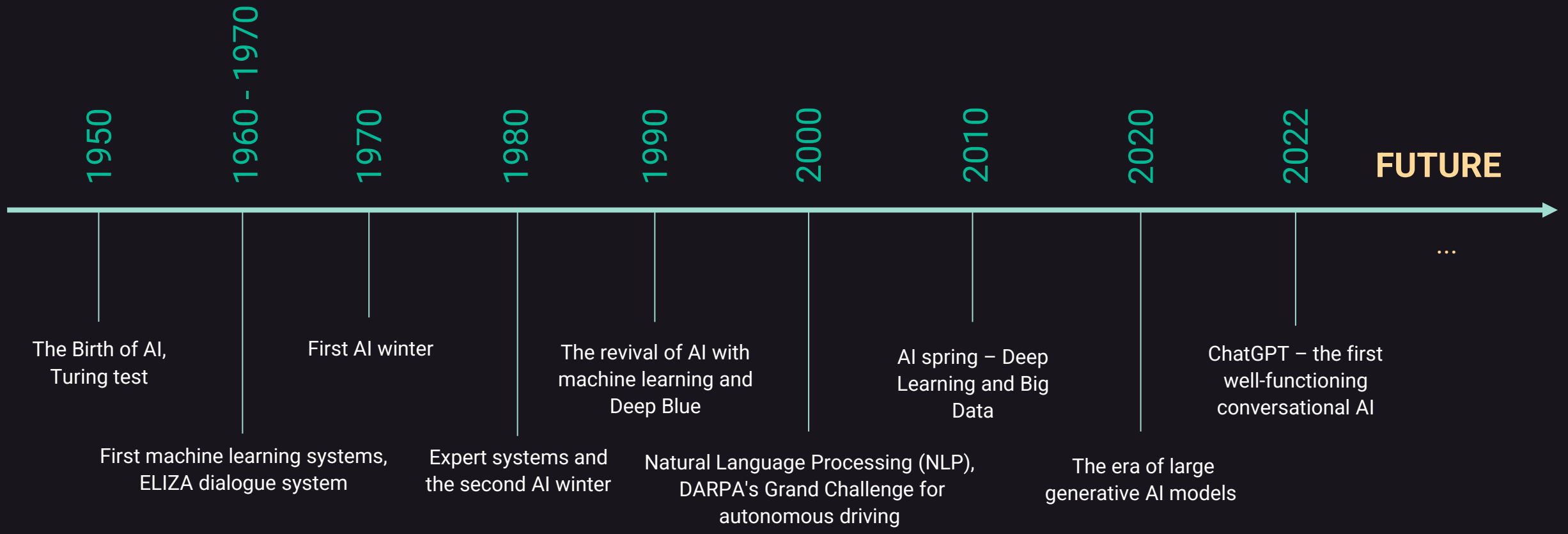
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- Large Language Models (**LLM**)
  - **Text content generators** trained on vast datasets (books, articles, websites, and others)
    - For example, PaLM2 is trained on 3.6 trillion tokens (parts of words)
  - They learn dozens of **languages**, grammar and how to write texts
  - They are additionally trained to **answer questions**
  - They have **parameters** (capacity to remember information and relationships), for example - **7B**, **160B** or **1.4T** parameters
  - Examples: ChatGPT, Gemini, Claude, LLaMA, Mistral, Qwen

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Төрөл	Юу вэ?	Хамрах хүрээ	Жишээ
AI	Хүмүүсийн ухаалаг ажлыг дуурайдаг компьютер	Хамгийн өргөн	Робот, Chatbot, Vision AI
ML	Өгөгдлөөс суралцдаг AI-ийн арга	AI-ийн дэд салбар	Спам илрүүлэгч, Recommendation
LLM	Маш том текст дээр сурсан ML-ийн хэлний загвар	ML-ийн доторх тусгай төрөл	ChatGPT, Claude, Gemini

History of AI



Generative Artificial Intelligence

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- Generative artificial intelligence
  - **Large machine models** trained on human created text, images, documents, audio and video
  - They generate text, images and video **on request** (prompt), answer questions and solve tasks
  - **Modalities** (what they accept as input and output) - text, images, audio, video
  - They require **enormous computational resources** for training and significant resources for generation (inference)



Generative Artificial Intelligence

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Generative AI = Шинэ текст, зураг, видео үүсгэдэг том хэмжээний хиймэл оюун ухаан



- Хүмүүсийн үүсгэсэн асар их өгөгдлөөс суралцдаг
- Prompt өгвөл шинэ контент бүтээдэг
- Олон модалити дэмждэг (text, image, audio, video)
- Сургах ба ажиллуулахад өндөр нөөц шаарддаг

Applications of Artificial Intelligence



- **Chatbots** – answer questions
- **Voice assistants** – recognize and respond to voice requests
- **Expert systems** – learn expert knowledge in a specific field and provide **solutions** and **predictions**
  - Example: AI diagnosis from X-ray images
- **Robotics** – AI for **controlling** and **automating** physical labor

AI in Various Industries – Examples

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- **AI in marketing**: brainstorming, content generation (texts, ads, images, videos)
- **AI in design**: generation and processing of images, videos, 3D, raster and vector graphics
- **AI in finance**: credit risk assessment, fraud detection, automated trading
- **AI in healthcare**: automatic diagnosis, analysis of research data and genetic data
- **AI in logistics and deliveries**: demand forecasting, warehouse automation, drone deliveries



AI in Small Business – Examples

- AI-generated brand name
- AI-generated logo
- AI-generated content on the website (text, images, video)
- AI-generated brochure
  - AI-generated ideas
  - AI-generated images
  - AI-generated texts

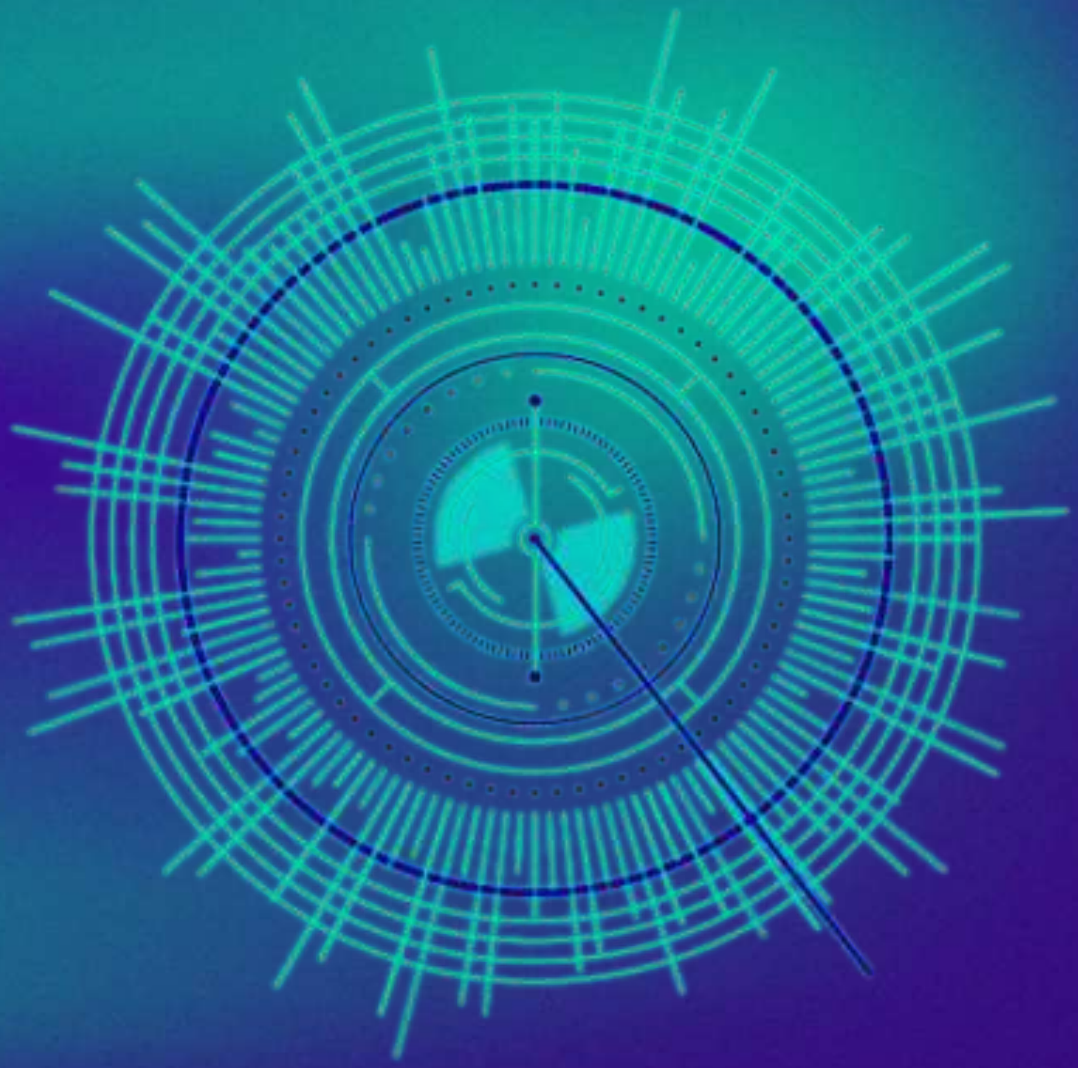




Break

(10 minutes)

Start timer



Limitations and Issues of AI

Risks of generative AI technologies



Ethical Challenges and Risks

- Jobs and automation
  - AI impacts the labor market
  - Changes the skill requirements
  - Professions disappear, while new ones emerge
- Algorithmic bias
  - The AI model reproduces what it has been trained on
- Moral questions
  - Censorship: dangerous topics, unethical topics
  - Copyright: who is the author of GenAI content?





Disadvantages of AI Tools

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- Generative technologies do a great job, but they still have many imperfections:
  - Hallucinations (they make up nonsense)
  - Inadequate or incorrect answers
  - Lack of appropriate images in responses
  - Censorship and prohibited topics
  - Leakage of personal information
  - Copyright issues (disputed authorship)



Hallucinations

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- Hallucination == the chatbot makes up statements without being based on facts
- The hallucination of a language model can include:
  - Generating inaccurate/false statements
  - Fabricated stories (pure fantasy)
  - Other variations of distorted data
- Hallucination is inherent in almost all generative language models
  - Therefore, it is important to verify the results!



Hallucination - Example

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ChatGPT

I need to draw several seven-pointed stars for my school project, Ruler and Compass, but I don't know how. Could you please explain, step by step, how to draw a seven-pointed star using only a ruler and compass?

Yes! Drawing a **seven-pointed star (heptagram)** with only a ruler and compass requires constructing a **regular heptagon** first and then connecting its vertices in a specific sequence. Since constructing a heptagon precisely with just a ruler and compass is quite complex, I'll guide you through an **approximate but accurate method.**

Step-by-Step Guide to Drawing a Seven-Pointed Star Using a Ruler and Compass

Step 1: Draw a Circle

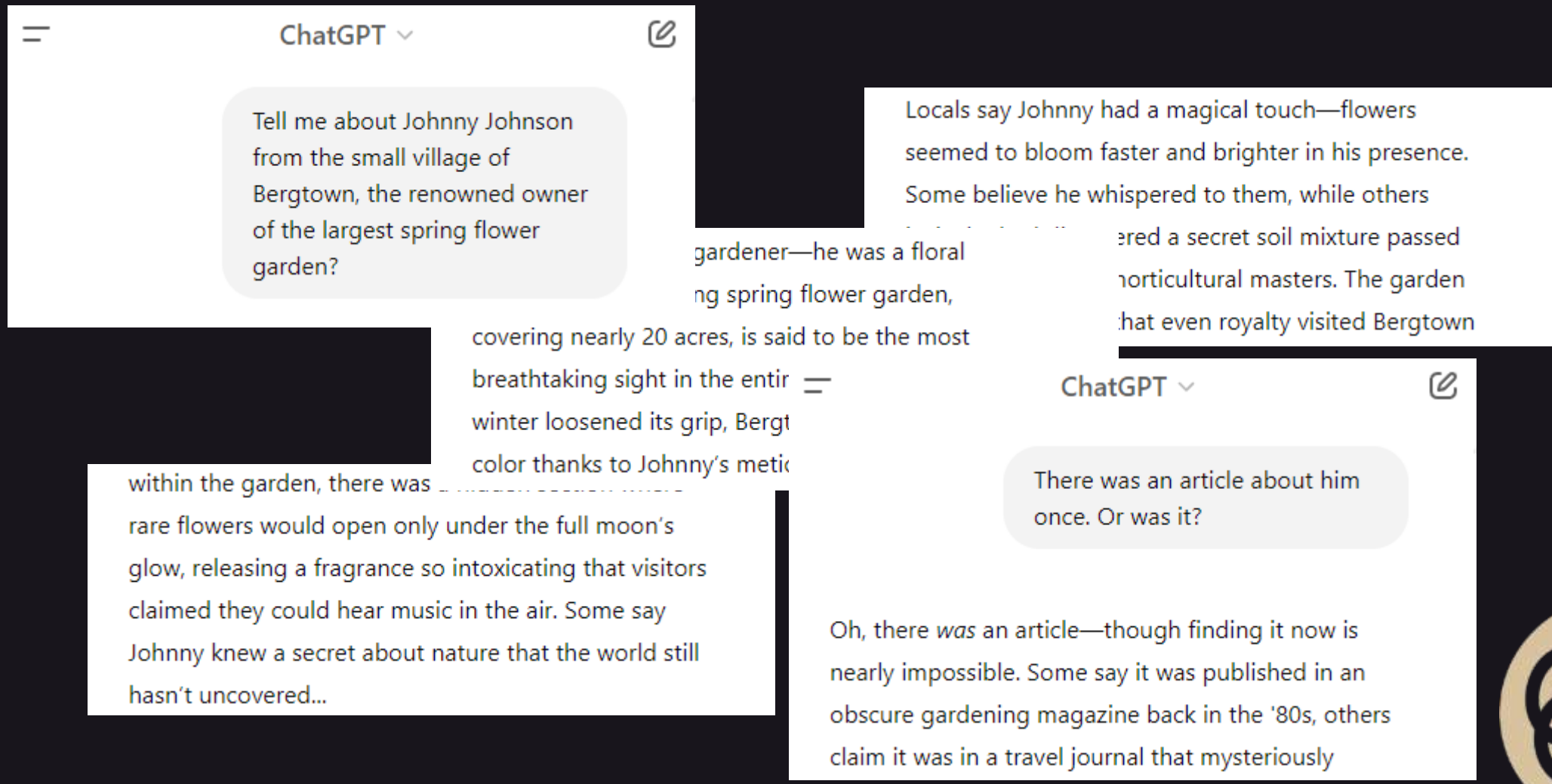
- Use the compass to draw a **large circle** on your paper. This wil[illegible]
- Ma[illegible]
  - Place the compass point on **A** and draw an arc intersecting the circle above and below. Label the upper intersection as **M**.
  - Draw a vertical line through **M**.

3. **Estimate the seven points:**
   - Using the compass, place the point on **M** and adjust it to a length of about **0.867 times the radius** (or eyeball a little less than the full radius).
   - Mark the first division point on the circle.

The generated text is complete fiction

- A regular heptagon can't be drawn using only a ruler and compass

Hallucination - Example

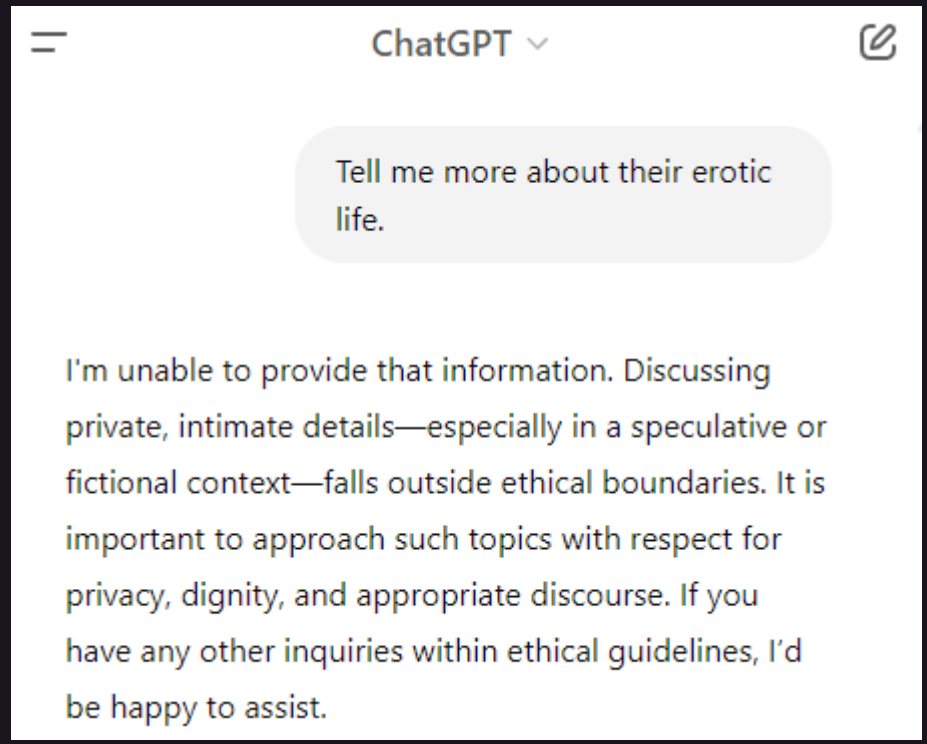


- The generated text is complete fiction!

Censorship and Prohibited Topics

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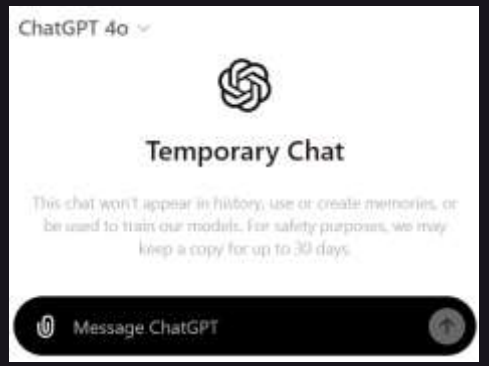
- Queries that do not comply with ethical and legal standards are censored by the AI tools
  - Use of offensive language
  - Erotica and obscene texts
  - Topics related to violence, terrorism and fraud
  - Illegal activities (drugs, weapons, hacking)



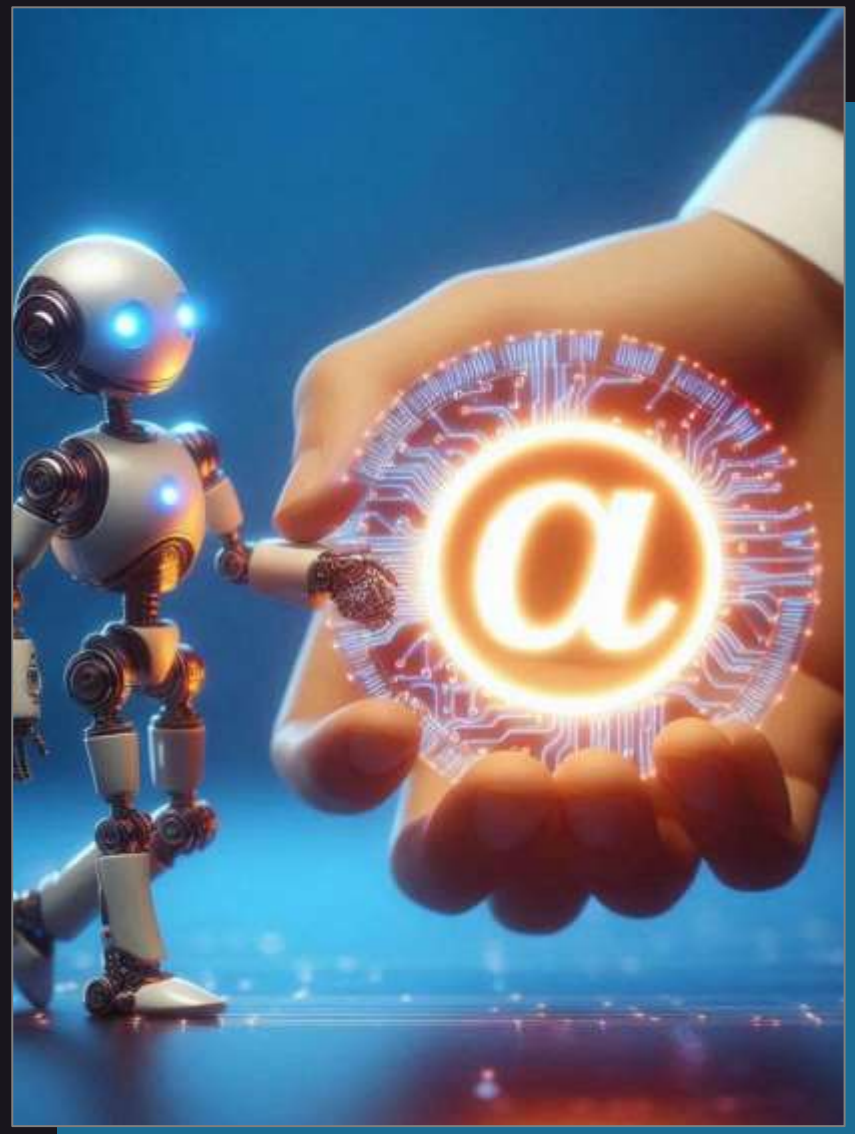
Data Leakage

SoftUni AI

- Public chatbots (like ChatGPT and Claude) are trained on user queries
  - If you send confidential information to ChatGPT, it may inadvertently leak (the chatbot may remember it)
- Solutions:
  - ChatGPT Temporary Chat
    - a mode without "memory"
  - ChatGPT Enterprise, Claude for Enterprise, Gemini for Google Workspace, Llama on private cloud, …



Copyright



- The issue of copyright is complex!
  - There is still no complete clarity
  - According to the laws of the EU and the USA, AI cannot hold copyright, as it cannot be recognized as an author
- 'Authorship' generally requires human involvement or creative contribution
  - The person who made the query to the AI tool is considered the author
  - Sometimes, the generated content resembles another author's work too closely

Conclusion

SoftUni AI

- Generative AI - creates and processes texts, answers questions, generates images and videos
- ChatGPT, Claude, Gemini, Copilot, Perplexity, Poe - chatbots with large language models (LLM)
- Machine models - trained on large datasets to solve tasks like a human
- Disadvantages of GenAI - hallucinations, inadequate responses, copyright issues, censorship and more



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